EPS 522/ENVS 423L Problem Set #5

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Calculations, methods, and assumptions should be spelled out so that your work can be reproduced. All work should be shown to receive full credit.

Upload a word document or PDF of your solutions/answers by midnight on Thursday, October 23 to Canvas.

1 Introduction

The purpose of this problem set is to reinforce your understanding of vadose zone processes.

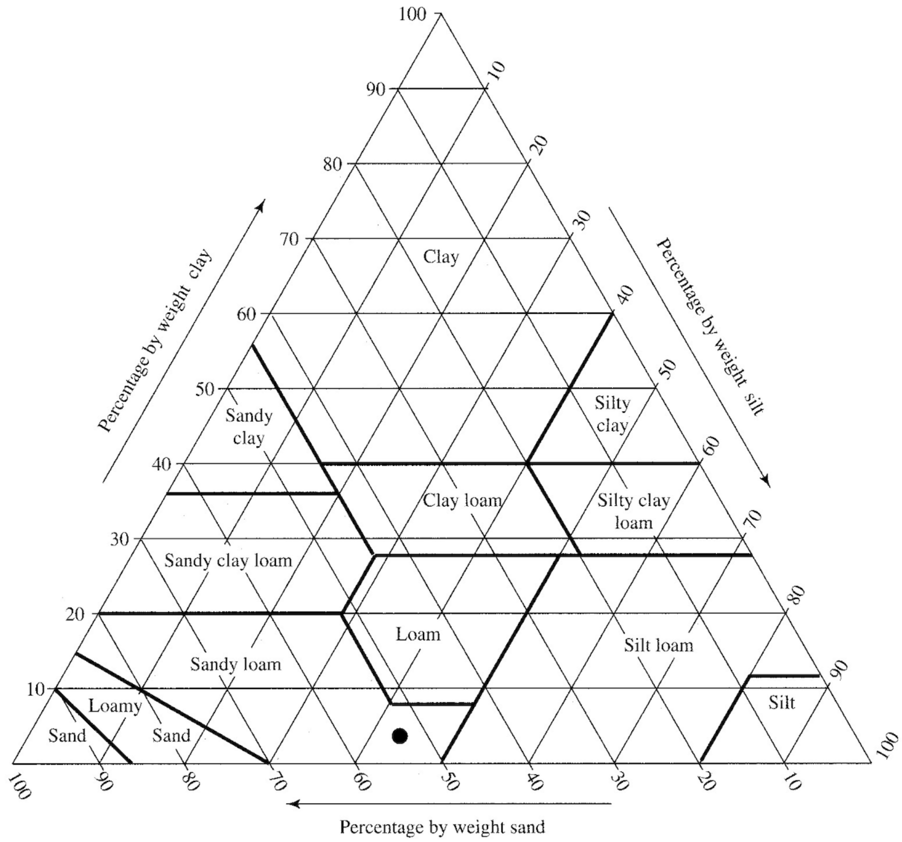
1.1 Grain Size Distribution and Soil Textural Class

Plot the grain size distribution for the following soils in the table below. Note that the Soil row (bolded values) is the grain diameter in mm (x-axis of grain size plot, log-transformed). The table values are the percent finer values (y-axis of grain size plot). (10 pts)

	Percent by Weight Finer Than Indicated Diameter (mm)									
Soil	**50**	**19**	**9.5**	**4.76**	**2**	**0.42**	**0.075**	**0.02**	**0.005**	**0.002**
1	100	100	100	100	100	100	97	79	45	16
2	100	100	98	94	70	19	15	8	3	2
3	93	91	88	85	69	44	40	27	13	6
4	100	100	100	100	100	97	92	75	47	31

1.1 Grain Size Distribution and Soil Textural Class, cont'd.

Determine the textural class of the 4 soils using the ternary plot below. Recall that these classes are the weight proportions of clay, silt, and sand after grains >2 mm (e.g., gravel) have been removed. (10 pts)



1.2 Properties

Field and oven-dry weights of four soil samples taken with a 10-cm long by 5-cm diameter cylindrical tube are given in the table below. Assume ρ_m=2.65 g/cm^3, calculate the field water content, degree of saturation, dry bulk density, and porosity of the soils. (20 pts)

Soil	Field Weight (g)	Oven-dry Weight (g)
1	302.5	264.8
2	376.3	308.0
3	422.6	388.6
4	468.3	441.7

Soil	ρ_b (g/cm^3)	ϕ (%)	θ (cm^3/cm^3)	S (%)
1				
2				
3				
4				

1.3 Tension

Consider two adjacent tensiometers inserted into an unsaturated sandy-loam soil to determine whether water flow is toward or away from the surface. Tensiometer A is placed at a depth of 20 cm; tensiometer B is at 60 cm. The following readings are obtained at different times. Which way is the water flowing (towards or away from surface) at each time? (15 pts)

	Time 1	Time 2	Time 3	Time 4	Time 5
ψ at A (cm)	-123	-106	-9	-211	-20
ψ at B (cm)	-22	-51	-65	-185	-6
Direction of Flow					

1.4 Moisture- and Unsaturated K-characteristic Curves

In the Vadose Zone pt2 lecture we saw two analytic approaches for the relation between $\psi - \theta$ and $K - \theta$, Campbell (1974) on slide 12 and van Genuchten (1980) on slide 27. Use the Campbell relation (given below) and the following table to plot the two characteristic curves for one of the soils in 1.2. (20 pts)

For $S < 1$,

$$|\psi(\theta)| = |\psi_{ae}| \cdot (\frac{\phi}{\theta})^b$$

For S=1,

$$|\psi(\theta)| = |\psi_{ae}|$$

$$K(\theta) = (\frac{\theta}{\phi})^c \cdot K^*$$

where,

$$c = 2 \cdot b + 3$$

Soil Texture	ϕ	K_h^* (cm s^{-1})	$\lvert\psi_{ae}\rvert$ (cm)	b
Sand	0.395 (0.056)	1.76×10^{-2}	12.1 (14.3)	4.05 (1.78)
Loamy sand	0.410 (0.068)	1.56×10^{-2}	9.0 (12.4)	4.38 (1.47)
Sandy loam	0.435 (0.086)	3.47×10^{-3}	21.8 (31.0)	4.90 (1.75)
Silt loam	0.485 (0.059)	7.20×10^{-4}	78.6 (51.2)	5.30 (1.96)
Loam	0.451 (0.078)	6.95×10^{-4}	47.8 (51.2)	5.39 (1.87)
Sandy clay loam	0.420 (0.059)	6.30×10^{-4}	29.9 (37.8)	7.12 (2.43)
Silty clay loam	0.477 (0.057)	1.70×10^{-4}	35.6 (37.8)	7.75 (2.77)
Clay loam	0.476 (0.053)	2.45×10^{-4}	63.0 (51.0)	8.52 (3.44)
Sandy clay	0.426 (0.057)	2.17×10^{-4}	15.3 (17.3)	10.4 (1.64)
Silty clay	0.492 (0.064)	1.03×10^{-4}	49.0 (62.1)	10.4 (4.45)
Clay	0.482 (0.050)	1.28×10^{-4}	40.5 (39.7)	11.4 (3.70)

1.5 Ring Infiltrometer

The results from a double ring infiltrometer test for two different sites are given in the below table. Plot the curves and report the implied saturated hydraulic conductivity for each. Based on the table in the previous problem set, what soil textures are A and B closest to? (20 points)

	Infiltration Rate (mm/hr)	
Time (min)	Soil A	Soil B
1	367.2	183.6
2	306	61.2
4	229.5	45.9
8	191.25	38.25
10	168.3	33.66
15	153	30.6
20	150.45	30.09
30	155.55	31.11
40	147.9	29.58
50	145.35	29.07
60	153	30.6